ASSIGNMENT No:-01

Write a Python program to plot a few activation functions that are being used in neural networks.

In [1]:
```
import numpy as np
import matplotlib.pyplot as plt
```

In [2]:
```
def sigmoid(x):
    return 1/(1+np.exp(-x))
```

In [3]:
```
def tanh(x):
    return np.tanh(x)
```

In [4]:
```
def relu(x):
    return np.maximum(0,x)
```

In [5]:
```
def leaky_relu(x,alpha=0.1):
    return np.where(x>0,x,alpha*x)
```

In [6]:
```
def softmax(x):
    e=np.exp(x-np.max(x))
    return e/np.sum(e)
```

In [7]:
```
inputs=np.array([-3,-1,0,1,-3])
print("input val",inputs)
print("sigmoid",sigmoid(inputs))
print("tanh",tanh(inputs))
print("relu",relu(inputs))
print("leaky_relu",leaky_relu(inputs))
print("softmax",softmax(inputs))
```

```
input val [-3 -1  0  1 -3]
sigmoid [0.04742587 0.26894142 0.5        0.73105858 0.04742587]
tanh [-0.99505475 -0.76159416  0.          0.76159416 -0.99505475]
relu [0 0 0 1 0]
leaky_relu [-0.3 -0.1  0.   1.  -0.3]
softmax [0.01189446 0.08788884 0.23890664 0.64941559 0.01189446]
```

In [8]:
```
x=np.linspace(-10,10,400)
plt.figure(figsize=(12,12))
plt.subplot(3,2,1)
plt.plot(sigmoid(x))
plt.title("sigmoid")
plt.xlabel("x")
plt.ylabel("f(x)")
plt.grid(True)
```

```
plt.subplot(3,2,2)
plt.plot(tanh(x))
plt.title("tanh")
plt.xlabel("x")
plt.ylabel("f(x)")
plt.grid(True)

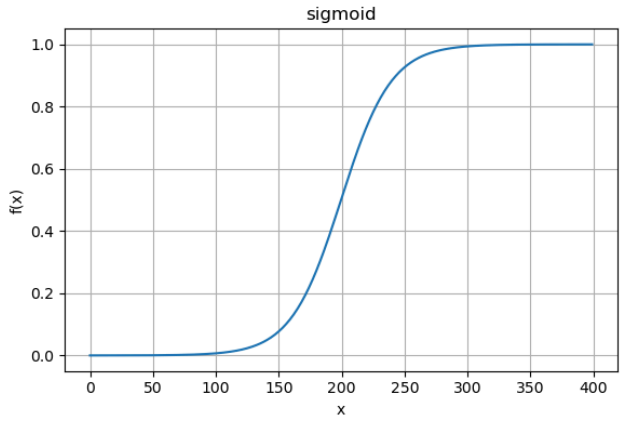
plt.subplot(3,2,3)
plt.plot(relu(x))
plt.title("relu")
plt.xlabel("x")
plt.ylabel("f(x)")
plt.grid(True)

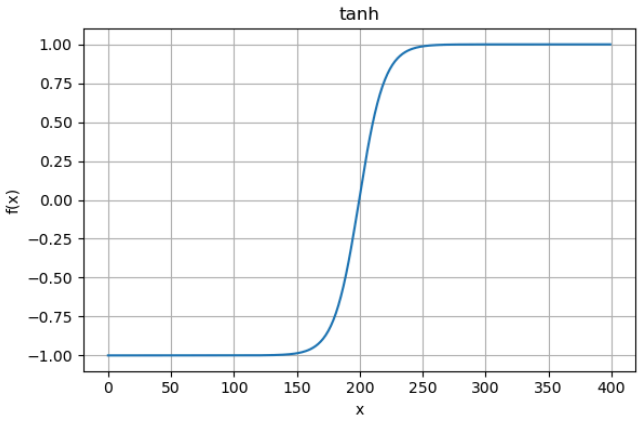
plt.subplot(3,2,4)
plt.plot(leaky_relu(x))
plt.title("leaky_relu")
plt.xlabel("x")
plt.ylabel("f(x)")
plt.grid(True)

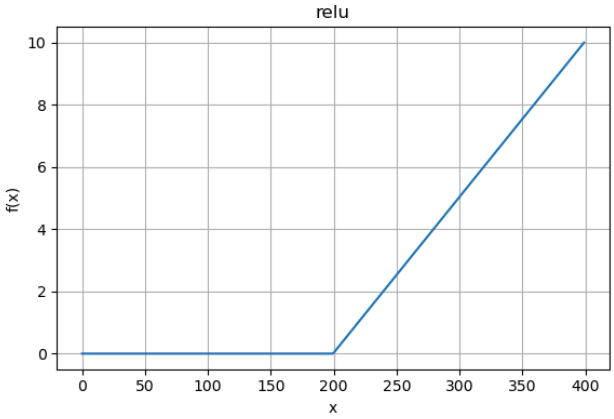
plt.subplot(3,2,5)
plt.plot(softmax(x))
plt.title("softmax")
plt.xlabel("x")
plt.ylabel("f(x)")
plt.grid(True)

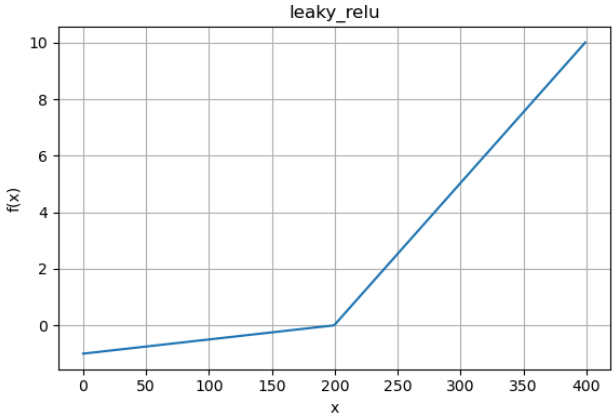
plt.tight_layout()

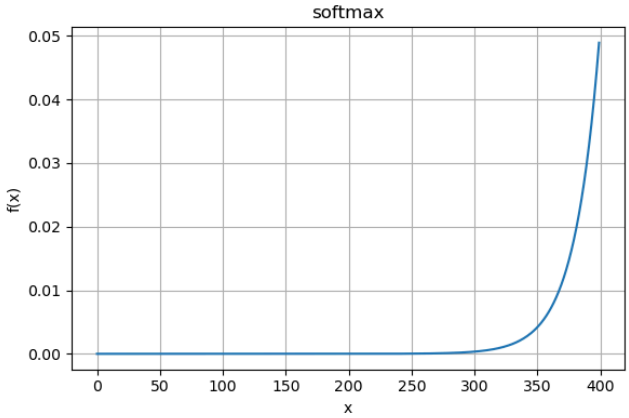
plt.show()
```











In [ ]: